

Binary Coded Decimal

BCD (1248)

=> Also called positively weighted code
=> BCD is a way to express each of the decimal digits with a binary code.
=> There are only 10 valid codes with 6 invalid codes as given below.

Decimal digit BCD (8421)

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

These are valid

These are invalid.....

10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Decimal to BCD

To convert any decimal number to BCD, simply replace each decimal digit with a valid 4-bit code.

Example:

Convert each of the following decimal numbers to BCD.

(a) 35

$$\begin{array}{cc} \overbrace{3} & \overbrace{5} \\ \underline{0011} & \underline{0101} \\ (35)_{10} = & (00110101)_{BCD} \end{array}$$

(b) 98

9
1001

8
1000

$(98)_{10} = (10011000)_{BCD}$

(c) 170

1
0001

7
0111

0
0000

$(170)_{10} = (00010111000)_{BCD}$

(d) 2469

2

4

6

9

$(2469)_{10} = (0010010001101001)_{BCD}$

BCD to Decimal

Example: Convert each of the following BCD to Decimal.

(a) 10000110

1000

8

0110

6

(d) 110010

0011 0010

32 = (32)₁₀

$(86)_{10} = 10000110$

(b) 001101010001

$= (351)_{10}$

(c) 1001000

0100 1000 = (48)₁₀

BCD Addition

Rules:

i) Perform addition like binary addition.

ii) if $\text{sum} \leq 9$, then it is a valid BCD number.

iii) if $\text{sum} > 9$ or a carry a carry out of a 4-bit group is generated, then an invalid result is generated.

iv) Add 6 (0110) to the 4-bit sum to skip the six invalid states.

v) if a carry results when 6 is added, simply add the carry to the next 4-bit.

Example: Add the following BCD numbers: (valid)

(a) 0011 + 0100

0011

3

0100

+4

0111

7

(b) $00100011 + 00010101$

$$\begin{array}{r} 00100011 \quad 23 \\ + 00010101 \quad + 15 \\ \hline 00111000 \quad 38 \end{array}$$

(c) $10000110 + 00010011$

$$\begin{array}{r} 10000110 \quad 86 \\ + 00010011 \quad + 13 \\ \hline 10011001 \quad 99 \end{array}$$

Example: Add the following BCD Numbers (Invalid).

(a) $1001 + 0100$

$$\begin{array}{r} 1001 \quad 9 \\ + 0100 \quad + 4 \\ \hline 1101 \quad 13 \\ \hline 10011 \end{array}$$

Invalid BCD number > 9

0001 0011

00010011 valid BCD number

(b) $1001 + 1001 = ?$

$$\begin{array}{r} 1001 \\ + 1001 \\ \hline 00010010 \end{array}$$

0001 0010

00010010 valid

(c) $00010110 + 00010101$

$$\begin{array}{r} 00010110 \\ + 00010101 \\ \hline 00101011 \rightarrow \text{invalid} > 9 \\ \uparrow 0110 \\ 00110001 \\ (00110001) \text{ Ans} \end{array}$$

(d)

$01100111 + 01010011$

$$\begin{array}{r} 01100111 \\ + 01010011 \\ \hline 10111010 \\ + 01100110 \\ \hline 10011000 \\ \hline 0010001 \\ \hline 0010 \\ 0001000000 \text{ Ans} \end{array}$$

Both groups are invalid add > 9

6 (0110)

Note BCD is a Self complementing code. Self complementing codes provide the 9's complement of a decimal number, just by interchanging 1's and 0's in its equivalent BCD representation.

Add 1001 + 1000

$$\begin{array}{r}
 1001 \quad 9 \\
 + 1000 \quad +8 \\
 \hline
 10001 \quad 17 \\
 \hline
 0110 \\
 \hline
 0111 \\
 \hline
 00010111 \text{ Ans}
 \end{array}$$

Add 1001 + 0111

$$\begin{array}{r}
 1001 \quad 9 \\
 + 0111 \quad +7 \\
 \hline
 10000 \quad 16 \\
 \hline
 0110 \\
 \hline
 0110 \\
 \hline
 00010110 \text{ Ans}
 \end{array}$$

(e)

00100101 + 00100111

$$\begin{array}{r}
 0010 \quad 0101 \quad 25 \\
 + 0010 \quad 0111 \quad 27 \\
 \hline
 0100 \quad 1100 \quad 52 \\
 \hline
 0101 \quad 1001 \quad 29
 \end{array}$$

01010010 Ans

52

Gray Code

Also called unit distance code.
 ⇒ The gray code is unweighted and non arithmetic code.

⇒ It exhibits only a single bit change from one code word to the next in sequence.

Decimal	Binary	Gray code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0101
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

from above adjacent numbers it is clear that there is only a single bit change in Gray coded numbers to the next in sequence. However binary numbers change appear more than one bit often.

Binary to - Gray code Conversion

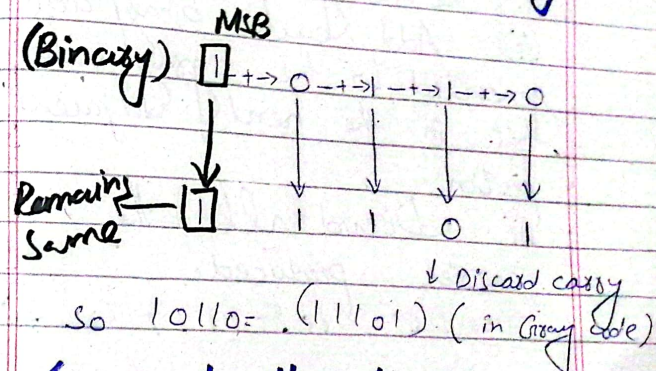
Rules:

- (i) The most significant bit (left most) in the Gray code is the same as the MSB in the binary number.
- (ii) Going from left to the right, added each adjacent pair of binary code bits to get the next Gray coded bit.

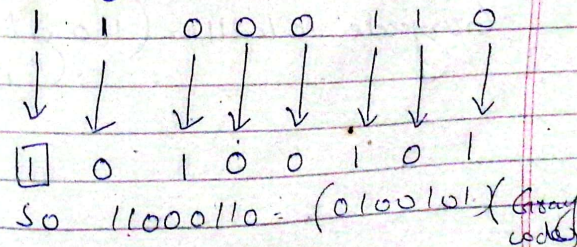
(iii) Discard all the carries produced.

Example:-

Convert the binary number 10110 to Gray code.



Convert the binary number 11000110 to Gray code.



Gray to binary Code:

Rules:

- (i) The most significant bit in the binary code is the same as the MSB in the Gray code.
- (ii) Add each binary code generated to the gray code bit in the next adjacent position.
- (iii) Discard all the carries produced.

Example: 10101111 to binary

(Gray) 1 0 1 0 1 1 1 1

↓ + ↓ + ↓ + ↓ + ↓ + ↓ + ↓

1 0 0 1 0 1 0

So

Graycode 10101111 = (11001010)
(in Binary Code)